

ACTL3142 Tutorial

Week 3: Multiple Linear Regression

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T1 2026

Recap from Last Week

From Week 2 (Simple Linear Regression):

- ▶ Model: $y = \beta_0 + \beta_1 x + \varepsilon$
- ▶ OLS: minimise RSS to get $\hat{\beta}_0, \hat{\beta}_1$
- ▶ Weak assumptions for point estimates; strong assumptions (normality) for inference
- ▶ ANOVA decomposition: $TSS = MSS + RSS$
- ▶ $R^2 = MSS/TSS$

Today: **multiple predictors**, model selection, and diagnostics.

Multiple Linear Regression: The Model

Extend SLR to p predictors:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \varepsilon$$

Interpreting the coefficients:

- ▶ β_j : average change in Y per unit increase in X_j , *holding all other predictors fixed*
- ▶ We are now fitting a hyperplane rather than a line

Partial vs total effect

In SLR, $\hat{\beta}_1$ captures the **total** association between X and Y . In MLR, $\hat{\beta}_j$ captures the **partial** effect of X_j after accounting for the other predictors. These can look quite different.

Matrix Notation and OLS

Write the model as $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, where $\mathbf{X}$ is $n \times (p + 1)$ with a leading column of 1's for the intercept.

The OLS estimator (minimising RSS):

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

Under the weak assumptions:

- ▶ Unbiased: $\mathbb{E}[\hat{\boldsymbol{\beta}} \mid \mathbf{X}] = \boldsymbol{\beta}$
- ▶ Variance-covariance: $\text{Var}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$
- ▶ We estimate σ^2 with $s^2 = \text{RSS} / (n - p - 1)$

The (j, j) diagonal of $(\mathbf{X}^\top \mathbf{X})^{-1}$ gives you $\text{Var}(\hat{\beta}_j) / \sigma^2$. This comes up in Q8.

Deriving the OLS Estimator

Minimise RSS in matrix form:

$$\text{RSS}(\beta) = (\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta)$$

Expand:

$$= \mathbf{y}^\top \mathbf{y} - 2\beta^\top \mathbf{X}^\top \mathbf{y} + \beta^\top \mathbf{X}^\top \mathbf{X} \beta$$

Differentiate w.r.t. β and set to zero:

$$\frac{\partial \text{RSS}}{\partial \beta} = -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X} \beta = \mathbf{0}$$

This gives the **normal equations** $\mathbf{X}^\top \mathbf{X} \hat{\beta} = \mathbf{X}^\top \mathbf{y}$, so:

$$\boxed{\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}}$$

This requires $\mathbf{X}^\top \mathbf{X}$ to be invertible. It won't be if predictors are perfectly collinear (e.g. including all dummy levels).

Hypothesis Tests in MLR

To test whether predictor X_j contributes to the model:

$$H_0: \beta_j = 0 \quad \text{vs} \quad H_1: \beta_j \neq 0$$

$$t_j = \frac{\hat{\beta}_j}{\widehat{SE}(\hat{\beta}_j)} \sim t_{n-p-1} \quad \text{under } H_0$$

R's `summary(lm(...))` reports the estimate, SE, t -value, and p -value for each coefficient.

A small p -value (e.g. < 0.05) suggests that predictor is significant *given the other predictors in the model*.

“Given the other predictors” matters. A variable can be significant in SLR but not in MLR (or vice versa) because of correlations between predictors.

Q10: Identifying Significant Predictors

★ Solution A regression model of a response Y on four independent variables (aka predictors) X_1, X_2, X_3 and X_4 has been fitted to a data consisting of 212 observations. The computer output from estimating this model is given below:

Regression Analysis

The regression equation is

$$y = 3894 - 50.3 x_1 + 0.0826 x_2 + 0.893 x_3 + 0.137 x_4$$

Predictor	Coef	SE Coef	T
Constant	3893.8	409.0	9.52
x1	-50.32	9.062	-5.55
x2	0.08258	0.02133	3.87
x3	0.89269	0.04744	18.82
x4	0.13677	0.05303	2.58

Using a level of significance of 5%, which of the following statement(s) is/are TRUE?

- All the predictor variables have a significant influence on Y .
- The variable X_2 is not a significant variable.
- The variable X_3 is not a significant variable.
- Exactly 3 of the 4 predictors are significant.

ANOVA Table for MLR

Source	D.F.	SS	MS	F
Regression	p	MSS	$\frac{\text{MSS}}{p}$	$\frac{\text{MSS}/p}{\text{RSS}/(n-p-1)}$
Error	$n - p - 1$	RSS	$\frac{\text{RSS}}{n - p - 1}$	
Total	$n - 1$	TSS		

Same structure as the SLR table from last week. The only change: Model D.F. = p instead of 1, Error D.F. = $n - p - 1$ instead of $n - 2$.

F-Test and Adjusted R^2

F-test for overall significance:

$$H_0: \beta_1 = \beta_2 = \cdots = \beta_p = 0$$

$$F = \frac{\text{MSS}/p}{\text{RSS}/(n-p-1)} \sim F_{p, n-p-1} \quad \text{under } H_0$$

Why not just use individual t-tests?

- ▶ Individual p-values can all be large while the model is jointly significant (or vice versa)
- ▶ With many predictors, some will look significant by chance

Adjusted R^2 :

$$R_{\text{adj}}^2 = 1 - \frac{\text{RSS}/(n-p-1)}{\text{TSS}/(n-1)}$$

Unlike R^2 , this can *decrease* when you add unhelpful predictors.

Q4 and Q5: Coefficient Interpretation and Adjusted R^2

★ [Solution](#) The following model was fitted to a sample of supermarkets in order to explain their profit levels:

4.
$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon$$

where

- y = profits, in thousands of dollars
- x_1 = food sales, in tens of thousands of dollars
- x_2 = nonfood sales, in tens of thousands of dollars, and
- x_3 = store size, in thousands of square feet.

The estimated regression coefficients are given below:

$$\hat{\beta}_1 = 0.027 \text{ and } \hat{\beta}_2 = -0.097 \text{ and } \hat{\beta}_3 = 0.525.$$

Which of the following is TRUE?

- a. A dollar increase in food sales increases profits by 2.7 cents.
- b. A 2.7 cent increase in food sales increases profits by a dollar.
- c. A 9.7 cent increase in nonfood sales decreases profits by a dollar.
- d. A dollar decrease in nonfood sales increases profits by 9.7 cents.
- e. An increase in store size by one square foot increases profits by 52.5 cents.

★ [Solution](#) In a regression model of three explanatory variables, twenty-five observations were used to calculate the least squares estimates. The total sum of squares and model sum of squares were found to be 666.98 and 610.48, respectively. Calculate the adjusted coefficient of determination (i.e., adjusted R^2).

5. a. 89.0%
- b. 89.4%
- c. 89.9%
- d. 90.3%
- e. 90.5%

Encoding Categorical Variables

A categorical predictor with q levels can't just be entered as a number; that would impose a false ordering.

Dummy encoding:

- ▶ Pick one level as the “base case” (absorbed into β_0)
- ▶ Create $q - 1$ binary (0/1) indicators for the remaining levels
- ▶ Each dummy coefficient $\hat{\beta}_j$ is the expected *difference* from the base category

Why drop one level?

With all q dummies plus an intercept, the columns of $\mathbf{X}$ sum to the intercept column (perfect collinearity). $(\mathbf{X}^\top \mathbf{X})$ becomes singular and OLS breaks.

Interaction and Polynomial Terms

The basic MLR model assumes effects are **additive**. Two common extensions:

Interaction terms:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \varepsilon$$

Rewrite: $Y = \beta_0 + (\beta_1 + \beta_3 X_2) X_1 + \beta_2 X_2 + \varepsilon$, so the effect of X_1 on Y now *depends on* X_2 .

Polynomial terms (handle non-linearity within the linear framework):

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \varepsilon$$

Still linear regression (linear in the β 's); X^2 is just another predictor.

If the interaction $X_1 X_2$ is significant, keep both X_1 and X_2 in the model even if their individual p-values are large (hierarchy principle).

Q8 and Q9: SE from $(\mathbf{X}^\top \mathbf{X})^{-1}$ and Dummy Interpretation

★ [Solution](#) You have information on 62 purchases of Ford automobiles. In particular, you have the amount paid for the car y in hundreds of dollars, the annual income of the individuals x_1 in hundreds of dollars, the sex of the purchaser (x_2 , 1=male and 0=female) and whether or not the purchaser graduated from college (x_3 , 1=yes, 0=no). After examining the data and other information available, you decide to use the regression model:

8.
$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon.$$

You are given that:

$$(\mathbf{X}^\top \mathbf{X})^{-1} = \begin{bmatrix} 0.109564 & -0.000115 & -0.035300 & -0.026804 \\ -0.000115 & 0.000001 & -0.000115 & -0.000091 \\ -0.035300 & -0.000115 & 0.102446 & 0.023971 \\ -0.026804 & -0.000091 & 0.023971 & 0.083184 \end{bmatrix}$$

and the mean square error for the model is $s^2 = 30106$. Calculate $\text{SE}(\hat{\beta}_2)$.

- a. 0.17
- b. 17.78
- c. 50.04
- d. 55.54
- e. 57.43

★ [Solution](#) Suppose in addition to the information in the previous question, you are given:

9.
$$\mathbf{X}^\top \mathbf{y} = \begin{bmatrix} 9\,558 \\ 4\,880\,937 \\ 7\,396 \\ 6\,552 \end{bmatrix}.$$

Calculate the expected difference in the amount spent to purchase a car between a person who graduated from college and another one who did not.

Possible answers:

- a. 233.5
- b. 1,604.3
- c. 2,195.3
- d. 4,920.6
- e. 6,472.1

Computing $SE(\hat{\beta}_j)$ from $(\mathbf{X}^\top \mathbf{X})^{-1}$

From the variance-covariance matrix:

$$\text{Var}(\hat{\beta}_j) = \sigma^2 \cdot [(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj}$$

Plug in $s^2 = \text{MSE} = \text{RSS}/(n - p - 1)$ for σ^2 :

$$\widehat{SE}(\hat{\beta}_j) = \sqrt{s^2 \cdot [(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj}}$$

For Q8: $s^2 = 30,106$ (given as MSE). The relevant diagonal entry of $(\mathbf{X}^\top \mathbf{X})^{-1}$ is the (3,3) element for β_3 . Plug in and take the square root.

For Q9: the coefficient on the “college graduate” indicator gives the expected difference in spending compared to a non-graduate.

Model Selection

With p potential predictors, which ones should we keep?

- ▶ R^2 always goes up with more predictors, so it can't guide selection on its own

Criteria that penalise complexity:

- ▶ $C_p = \frac{1}{n}(\text{RSS} + 2d\hat{\sigma}^2)$ (lower is better)
- ▶ $\text{AIC} = 2d - 2\ln(\hat{L})$ (lower is better)
- ▶ $\text{BIC} = d\ln(n) - 2\ln(\hat{L})$ (lower is better)
- ▶ Adjusted R^2 (higher is better)

All four trade off fit vs complexity. BIC penalises harder than AIC when $n > 7$ (since $\ln(n) > 2$), so it tends to pick simpler models.

Subset Selection: Best Subset, Forward, Backward

Best subset selection:

1. For each $k = 0, 1, \dots, p$: fit all $\binom{p}{k}$ models with k predictors; pick best by R^2
2. Select the overall best $\mathcal{M}_k$ using C_p , BIC, or adjusted R^2

Catch: 2^p total models. Fine for small p , infeasible for large p .

Forward stepwise: start with no predictors; at each step add the one that most improves R^2 .

Backward stepwise: start with all predictors; at each step drop the one whose removal hurts R^2 least.

Both are greedy: they look one step ahead, so they don't guarantee the global best. But they're fast. With $p = 20$, best subset fits $\sim 1M$ models; forward fits ~ 211 .

Training Error vs Test Error

- ▶ **Training RSS** always decreases (or stays flat) as you add predictors
- ▶ **Test RSS** follows a U-shape: drops initially, then rises as the model overfits

Back to the bias-variance tradeoff from Week 1:

- ▶ More predictors $\Rightarrow$ less bias, more variance
- ▶ Fewer predictors $\Rightarrow$ more bias, less variance

Keep this in mind for Q2 and Q14

Training RSS tells you how well the model fits *its own* data. Test RSS tells you how well it **generalises**. They can tell very different stories.

Q2: True Linear vs Cubic Model

2.
 - a. Suppose that the true relationship between X and Y is linear, i.e. $Y = \beta_0 + \beta_1 X + \epsilon$. Consider the training residual sum of squares (RSS) for the linear regression, and also the training RSS for the cubic regression. Would we expect one to be lower than the other, would we expect them to be the same, or is there not enough information to tell? Justify your answer.
 - b. Answer (a) using test rather than training RSS.
 - c. Suppose that the true relationship between X and Y is not linear, but we don't know how far it is from linear. Consider the training RSS for the linear regression, and also the training RSS for the cubic regression. Would we expect one to be lower than the other, would we expect them to be the same, or is there not enough information to tell? Justify your answer.
 - d. Answer (c) using test rather than training RSS.

Solution

3.
 - a. Write out the design matrix $\mathbf{X}$ in the special case of the **simple linear regression model** (i.e., one predictor).
 - b. Write out the matrix $\mathbf{X}^\top \mathbf{X}$ for the simple linear regression model.
 - c. Write out the vector $\mathbf{X}^\top \mathbf{y}$ for the simple linear regression model.
 - d. Write out the matrix $(\mathbf{X}^\top \mathbf{X})^{-1}$ for the simple linear regression model.
 - e. Calculate $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ using your results above.

Where $\mathbf{y}$ is the vector of the response variable and $\hat{\boldsymbol{\beta}}$ is the vector of coefficients.

Q14: Stepwise Selection

★ [Solution](#) (ISLR2, Q6.1) We perform best subset, forward stepwise, and backward stepwise selection on a single data set. For each approach, we obtain $p + 1$ models, containing $0, 1, 2, \dots, p$ predictors. Explain your answers:

14. a. Which of the three models with k predictors has the smallest *training* RSS?
- b. Which of the three models with k predictors has the smallest *test* RSS?
- c. True or False:
- i. The predictors in the k -variable model identified by forward stepwise are a subset of the predictors in the $(k + 1)$ -variable model identified by forward stepwise selection.
 - ii. The predictors in the k -variable model identified by backward stepwise are a subset of the predictors in the $(k + 1)$ -variable model identified by backward stepwise selection
 - iii. The predictors in the k -variable model identified by backward stepwise are a subset of the predictors in the $(k + 1)$ -variable model identified by forward stepwise selection.
 - iv. The predictors in the k -variable model identified by forward stepwise are a subset of the predictors in the $(k + 1)$ -variable model identified by backward stepwise selection.
 - v. The predictors in the k -variable model identified by best subset are a subset of the predictors in the $(k + 1)$ -variable model identified by best subset selection.

Potential Problems in Linear Regression

1. **Non-linearity:** curved patterns in residual plots; try polynomial terms or transformations
2. **Correlated errors:** systematic patterns in residuals vs order/time
3. **Heteroscedasticity:** funnel shape in residuals; try log-transforming Y
4. **Outliers:** unusually large residuals; inspect with studentised residuals
5. **High leverage:** unusual predictor values; flag when $h_{ii} > 2(p+1)/n$
6. **Collinearity:** inflated SEs; $VIF(\hat{\beta}_j) = 1/(1 - R_{X_j|X_{-j}}^2)$, worry if $VIF > 5$
7. **Confounders:** a hidden variable driving both X and Y ; correlation $\neq$ causation

When diagnosing a model, start with the residual plot. Issues 1, 3, and 4 all show up there.

Individual t -Tests vs Overall F -Test

	Individual t -test	Overall F -test
Tests	$H_0 : \beta_j = 0$ (one predictor)	$H_0 : \beta_1 = \dots = \beta_p = 0$ (all)
Stat	$t_j = \hat{\beta}_j / \widehat{SE}(\hat{\beta}_j)$	$F = \frac{MSS/p}{RSS/(n-p-1)}$
Dist	t_{n-p-1}	$F_{p, n-p-1}$
Tells you	Does X_j add value <i>given the others</i> ?	Is the model better than just predicting $\bar{y}$?

In SLR ($p = 1$), these are equivalent: $F = t^2$. In MLR they answer different questions.

Thanks for coming! See you next week.

